

Zach Fisher

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LLM post-training at Google DeepMind, focused on RL and evaluation for agentic tool use and scientific reasoning. Previously a theoretical physicist: co-formulated the Quantum Null Energy Condition and co-authored its first proof.

EMPLOYMENT

Google DeepMind (formerly Google Research)

Staff Software Engineer

Senior Software Engineer

Software Engineer

Mountain View, CA

May 2025 – Present

April 2022 – May 2025

April 2020 – April 2022

- Built reward models for reinforcement-learning post-training of frontier LLMs: rubric-conditioned critiques, code-based auto-raters, and critic-driven self-improvement. These methods shipped in multiple Gemini training revisions.
- Conceived and led a scientific-agent benchmark for long-horizon optimization against real-world simulators, and used it to identify an exploration gap separating leading frontier model families. Collected hundreds of tasks across tens of engineering and science domains, and built the evaluation infrastructure and leaderboard. Now a standard Gemini internal leaderboard.
- Designed and scaled a failure-mode-driven synthetic data pipeline that generated targeted training examples at scale for SQL, data science, mathematical reasoning, and scientific coding, with automated multi-stage quality filtering and simulator-backed verification.
- Led research on simulator-driven RL for agentic tool use, building a procedural generator that produced dense-reward inverse problems across 5 scientific domains for RL post-training.
- Diagnosed root causes of frontier training-run failures, including silent infrastructure faults, reward-signal degradation, evaluation flakiness, and distribution shift in model behavior, and built the real-time health dashboards and sample-exploration tools used to triage live runs.
- Led attribute extraction for Google Shopping and Maps: tens of millions of edits across 80+ attributes at over 96% precision, with $3\text{--}5\times$ recall gains in four languages.
- Contributed to the Google I/O 2023 launch of Shopping Companion, standing up a 340B-parameter model to batch-generate the structured product comparisons (STRUM) behind the launch.
- Set technical direction beyond the immediate team: models, evaluation frameworks, and infrastructure adopted by partner groups in Search, Maps, Shopping, Travel, and post-training. Supported engineers and researchers across 6+ teams, and served as a teaching assistant for Alphabet's six-week Probability and Statistics for Machine Learning course.

Perimeter Institute for Theoretical Physics

Postdoctoral Researcher

Waterloo, ON, Canada

September 2017 – February 2020

- It from Qubit fellow in quantum field theory, quantum information, and quantum gravity.
- Characterized how information escapes evaporating black holes coupled to external thermal baths, and derived entropy bounds and quantum energy conditions linking quantum information inequalities to spacetime geometry.

EDUCATION

University of California, Berkeley

August 2012 – May 2017

Ph.D. in Physics. Thesis: "Entropy Bounds and Entanglement." Co-formulated the Quantum Null Energy Condition and Quantum Focusing Conjecture, co-authored the QNEC's first proof, and proved a quantum generalization of the Bousso entropy bound.

Massachusetts Institute of Technology

September 2008 – June 2012

S.B. in Physics.

SKILLS

- **Research:** RL post-training, RLHF, reward and value modeling, agent evaluation, tool use, synthetic data generation, long-context architectures, mathematical reasoning, scientific AI.
- **Engineering:** Python, JAX, PyTorch, TensorFlow, C++, SQL, Apache Beam, large-scale distributed training.

SELECTED PUBLICATIONS

Machine Learning

- Gemini Team, Google (**Z. Fisher** et al.). [“Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities.”](#) arXiv:2507.06261 (2025). 4,200+ citations.
- J. Ainslie, S. Ontanon, C. Alberti, V. Cvicek, **Z. Fisher** et al. [“ETC: Encoding Long and Structured Inputs in Transformers.”](#) EMNLP (2020). 520+ citations.
- S. Ontanon, J. Ainslie, **Z. Fisher**, V. Cvicek. [“Making Transformers Solve Compositional Tasks.”](#) ACL (2022). 120+ citations.
- R. Aksitov, S. Miryoosefi, **Z. Fisher** et al. [“ReST meets ReAct: Self-Improvement for Multi-Step Reasoning LLM Agent.”](#) arXiv:2312.10003 (2023). 95+ citations.
- S. Ontanon, J. Ainslie, V. Cvicek, **Z. Fisher**. [“LogicInference: A New Dataset for Teaching Logical Inference to Seq2Seq Models.”](#) arXiv:2203.15099 (2022).
- Y. Sheng, S. Gandhe, B. Kanagal, **Z. Fisher** et al. [“Measuring an LLM’s Proficiency at Using APIs: A Query Generation Strategy.”](#) KDD-ADS (2024).
- B. Gunel, J.B. Wendt, J. Xie, **Z. Fisher** et al. [“STRUM-LLM: Attributed and Structured Contrastive Summarization.”](#) arXiv:2403.19710 (2024).

Theoretical Physics

- R. Bousso, **Z. Fisher**, S. Leichenauer, A. Wall. [“Quantum Focusing Conjecture.”](#) Phys. Rev. D 93, 064044 (2016). 450+ citations.
- R. Bousso, **Z. Fisher**, J. Koeller, S. Leichenauer, A. Wall. [“Proof of the Quantum Null Energy Condition.”](#) Phys. Rev. D 93, 024017 (2016). 280+ citations; *Editors’ Suggestion*.
- H.Z. Chen, **Z. Fisher**, J. Hernandez, R.C. Myers, S.M. Ruan. [“Information Flow in Black Hole Evaporation.”](#) JHEP 2020 (3), 152. 175+ citations.
- R. Bousso, H. Casini, **Z. Fisher**, J. Maldacena. [“Proof of a Quantum Bousso Bound.”](#) Phys. Rev. D 90, 044002 (2014). 160+ citations.
- R. Bousso, H. Casini, **Z. Fisher**, J. Maldacena. [“Entropy on a Null Surface for Interacting Quantum Field Theories and the Bousso Bound.”](#) Phys. Rev. D 91, 084030 (2015). 145+ citations.
- H.Z. Chen, **Z. Fisher**, J. Hernandez, R.C. Myers, S.M. Ruan. [“Evaporating Black Holes Coupled to a Thermal Bath.”](#) JHEP 2021 (1), 65. 115+ citations.

PATENTS AND DISCLOSURES

- A. Sadr, T. Yu, D. Li, **Z. Fisher** et al. [“Artificial Intelligence Generated Badges for Search.”](#) U.S. Patent Application 18/636,844 (2024).
- H. Wu, **Z. Fisher** et al. [“Automated Optimization of Structured Large Language Model Prompts via Iterative Evaluation.”](#) Technical Disclosure Commons (2026).

AWARDS

- Product Achievement Award, Google Research (2022).
- Publication and Open Source Excellence Award, Google Research (2021).
- It from Qubit Postdoctoral Fellowship, Simons Collaboration (2017).